# Case Study 53: Food Waste Monitoring System

## 1. Problem Statement

Compare prepared meals, served meals, and unserved leftovers across several days, calculate daily food waste percentages and monetary losses, and identify days that exceed acceptable wastage thresholds.

## 2. Objectives

- Track multi-day mess/canteen meal preparation and consumption metrics.
- Calculate exact unserved leftover percentages relative to total prepared quantities.
- Flag days exceeding acceptable wastage limits and compute cumulative monetary loss.

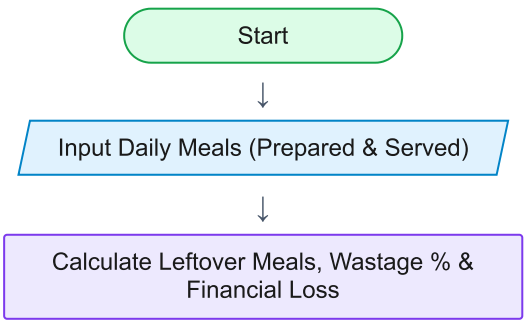
## 3. Assumptions

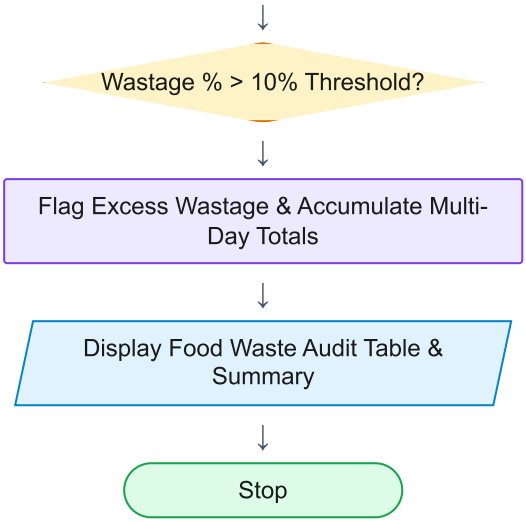
- Acceptable daily wastage threshold is **10.0%**.
- Cost per prepared meal unit is **Rs. 50.00**.
- Wastage Quantity = `Prepared Meals - Served Meals`.

## 4. Algorithm

1. Start.
2. Read daily log entries `[(day_name, prepared, served), ...]`.
3. Set `threshold_pct = 10.0` and `unit_cost = 50.0`.
4. For each day's record:
   - Calculate `leftover = prepared - served`.
   - Calculate `waste_pct = (leftover / prepared) * 100.0`.
   - Calculate `loss_cost = leftover * unit_cost`.
   - If `waste_pct > threshold_pct`, flag day as **EXCESS WASTAGE**.
   - Else flag day as **ACCEPTABLE**.
5. Compute cumulative total prepared, total leftover, overall waste percentage, and net financial loss.
6. Display daily monitoring table and audit summary.
7. Stop.

## 5. Flowchart






## 6. Pseudocode

```
START INPUT daily_records threshold = 10.0 unit_cost = 50.0 total_prep = 0 total_waste = 0 total_loss = 0.0 FOR EACH rec IN daily_records DO day = rec.day prep = rec.prepared served = rec.served waste = prep - served pct = (waste / prep) * 100.0 loss = waste * unit_cost total_prep = total_prep + prep total_waste = total_waste + waste total_loss = total_loss + loss IF pct > threshold THEN status = "EXCESS WASTAGE" ELSE status = "ACCEPTABLE" ENDIF PRINT day, prep, served, waste, pct, loss, status ENDFOR overall_pct = (total_waste / total_prep) * 100.0 PRINT total_prep, total_waste, overall_pct, total_loss STOP
```

## 7. Complete Python Source Code

```
      # Food Waste Monitoring System def monitor_food_waste(daily_logs,
cost_per_meal=50.0, max_allowed_pct=10.0): total_prepared = 0 total_waste = 0
total_financial_loss = 0.0 excess_days = [] print("=" * 74) print(" CANTEEN FOOD
WASTE AUDIT REPORT") print("=" * 74) print(f"{'Day':<12}{'Prepared':<10}{'Served':
<10}{'Waste(Meals)':<14}{'Waste %':<10}{'Loss (Rs.)':<12}{'Status':<10}") print("-"
* 74) for day, prepared, served in daily_logs: waste = prepared - served waste_pct
= (waste / prepared) * 100.0 loss = waste * cost_per_meal total_prepared +=
prepared total_waste += waste total_financial_loss += loss if waste_pct >
max_allowed_pct: status = "EXCESS" excess_days.append(day) else: status = "OK"
print(f"{day:<12}{prepared:<10}{served:<10}{waste:<14}{waste_pct:<10.1f}{loss:
<12,.2f}{status:<10}") overall_pct = (total_waste / total_prepared) * 100.0
print("-" * 74) print(f"Total Meals Prepared : {total_prepared}") print(f"Total
Meals Wasted : {total_waste} ({overall_pct:.2f}% of total production)")
print(f"Total Financial Loss : Rs. {total_financial_loss:,.2f}") print(f"Days
Exceeding {max_allowed_pct}%: {', '.join(excess_days) if excess_days else 'None'}")
print("=" * 74) # Multi-Day Canteen Data: (Day, Prepared Meals, Served Meals)
weekly_data = [ ("Monday", 500, 460), # 40 waste (8.0%) ("Tuesday", 550, 470), # 80
waste (14.5%) - EXCESS ("Wednesday", 520, 490), # 30 waste (5.8%) ("Thursday", 600,
510), # 90 waste (15.0%) - EXCESS ("Friday", 580, 540) # 40 waste (6.9%) ]
monitor_food_waste(weekly_data)
```

## 8. Test Case and Expected Output

**Evaluation Output for 5-Day Audit:**

```
      ============================================================================
CANTEEN FOOD WASTE AUDIT REPORT
============================================================================ Day
Prepared Served Waste(Meals) Waste % Loss (Rs.) Status ----------------------------
------------------------------------------- Monday 500 460 40 8.0 2,000.00 OK
Tuesday 550 470 80 14.5 4,000.00 EXCESS Wednesday 520 490 30 5.8 1,500.00 OK
Thursday 600 510 90 15.0 4,500.00 EXCESS Friday 580 540 40 6.9 2,000.00 OK --------
------------------------------------------------------------------ Total Meals
Prepared : 2750 Total Meals Wasted : 280 (10.18% of total production) Total
Financial Loss : Rs. 14,000.00 Days Exceeding 10.0%: Tuesday, Thursday
============================================================================
```

## 9. Conclusion

The Food Waste Monitoring System tracks canteen preparation efficiency, detects daily unserved meal wastage thresholds, and calculates the exact financial losses incurred, enabling institutions to optimize meal production and minimize food waste.